Reverse Words In String

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The problem

- Reverse the order of space-separated words, preserving all whitespace runs.

Intuition

- Reverse the order of words while keeping whitespace runs intact.

Approach

- 1. Tokenise into words and spaces
- 2. Reverse the token list
- 3. Join back

Java solution

```java
public class ReverseWordsInString {
    public static String reverseWordsInString(String s) {
        StringBuilder out = new StringBuilder();
        int end = s.length();
        for (int i = s.length() - 1; i >= -1; i--) {
            if (i == -1 || (s.charAt(i) == ' ') != (s.charAt(end - 1) == ' ')) {
                out.append(s, i + 1, end);
                end = i + 1;
            }
            if (end == 0) break;
        }
        return out.toString();
    }
}
```

Complexity

- Time: O(n)
- Space: O(n)

Recap

- Reverse the order of words while keeping whitespace runs intact.
- Complexity: O(n)